

Crank Arm Optimization

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1.Introduction

This report contains the design and development bicycle crank arm through the design stages of initial concept, simulation, optimization and final design. The design process follows Archer's design structure which ensures that all functional, structural, environmental and ergonomic requirements are met. The optimized mode is kept under 300g and has a minimum safety factor of 2.

2.Specification

The bicycle crank must be designed to meet the clients specification which is given in Table 1. The crank must meet the functional, structural and environmental requirements.

Table 1: Bike crank design specification

Parameter	Requirement	Achieved Yes/No	Evidence
Performance	Client weight is 70 kg; Max component weight is < 300 g		
Size	Max footprint detailed in Fig. 2		
Maintenance	No maintenance during lifespan		
Finish	Corrosion resistant		
Materials	Easily recyclable		
Transportable	Not easily damaged by impact		
Aesthetics	Must present an image of high-end performance		
Customer	BMX enthusiasts		
Safety	No sharp protrusions; safety factor of 2 implemented from design loading		
Environment	-10 °C to +40 °C, water and dust resistant		
Ergonomics	Compatible with existing mounts and pedals		

3.Analysis

This stage demonstrates understanding of the design problem, constraints and loading.

1. Client Requirements

The client requires a crank that is lightweight (less than 300 g) and strong enough to carry a 70 kg rider. The crank must maintain a safety factor of at least 2, resist corrosion, survive temperatures from -10 °C to +40 °C and has to be recyclable.

2. Design Constraints

Design envelope:

Figure 2 shows the maximum permissible volume (grey areas). The red regions (crank shaft attachment and pedal mounting holes) cannot be changed.

Mass limit:

The optimized crank must be less than 300 g. The initial CAD model has a mass of 727 g.

Safety factor:

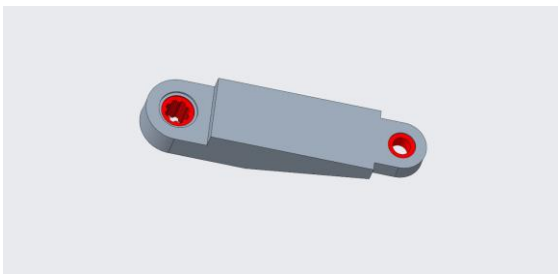
A minimum safety factor of 2 is required.
$$\text{FoS} = \frac{\sigma_{\text{yield}}}{\sigma_{\text{max, von Mises}}}$$

For the chosen material (Al 7075, $\sigma_{\text{yield}} = 498 \text{ MPa}$)

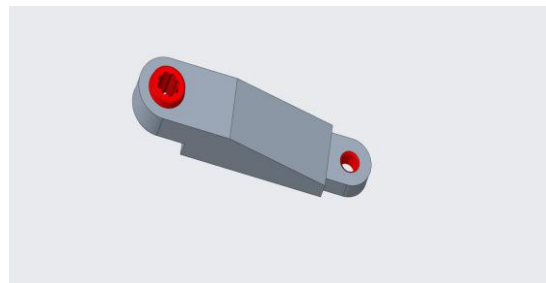
$$\sigma_{\text{allow}} = \frac{498 \text{ MPa}}{2} = 249 \text{ MPa}.$$

the final model should be $\sigma_{\text{max}} \leq 249 \text{ MPa}$.

3. Optimizing the Design



Front view



Back view

Figure 1 : Initial CAD design : (a) front view, (b) back view.

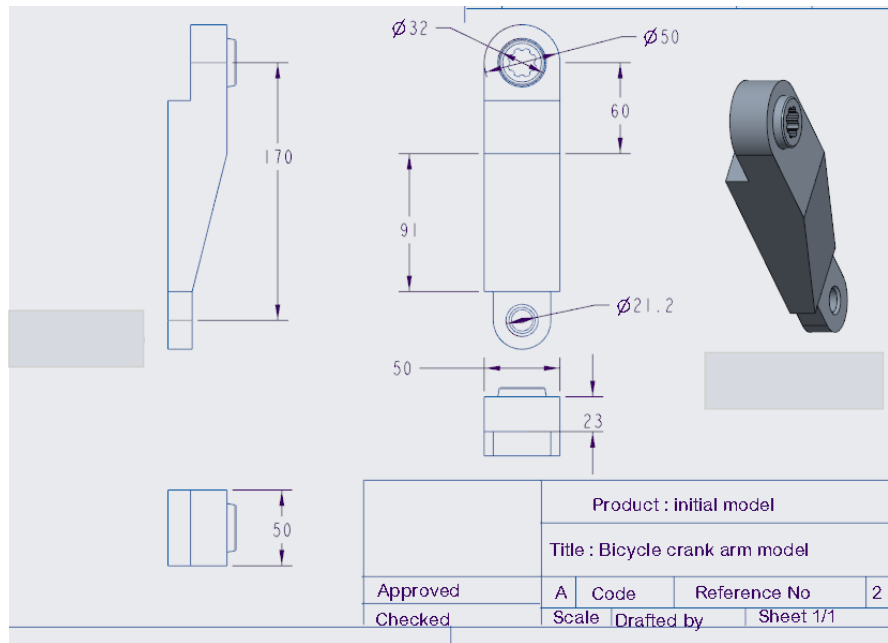


Figure 2 : 2D engineering drawing of the working.

VOLUME = 2.6043123e+05 MM³
 SURFACE AREA = 3.5529580e+04 MM²
 DENSITY = 2.7935544e-09 TONNE / MM³
 MASS = 7.2752880e-04 TONNE

Figure 3 : Initial mass measurement from CAD (727g)

4. Material Selection

The material chosen for the crank arm is Aluminum Alloy 7075 (Al 7075). This material was selected due to its excellent strength to weight ratio, good corrosion resistance and is also recyclable which aligns with the client specification.

Table 2 : Material properties of Al 7075

Property	Value
Density	2800 kg/m ³
Young's Modulus	73.05 GPa
Poisson's Ratio	0.33
Yield Strength	498 MPa
Corrosion resistance (Water)	Excellent
Fatigue strength (10 ⁷ cycles)	147 MPa

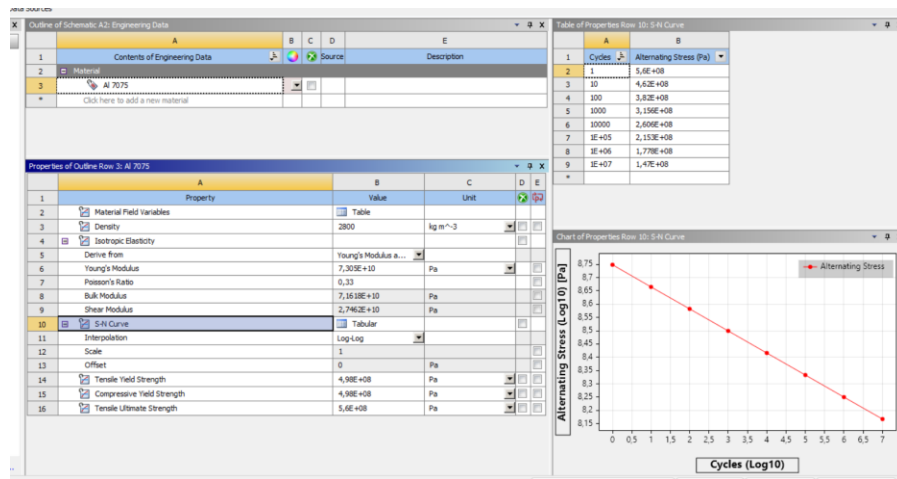


Figure 4 : ANSYS Engineering Data.

5. Load Calculation and Assumptions

For finite element analysis the loads acting on the crank arm is based on the client's rider mass of 70 kg.

Assumptions

The rider's full weight is assumed to act on one pedal at the position where the crank arm is horizontal and the pedal is the farthest distance from the crank. This represents the maximum bending moment.

The gravitational acceleration is taken as $g = 9.81 \text{ m/s}^2$.

The pedal lever arm length is $d = 0.17 \text{ m}$.

The crank shaft is fixed in all degrees of freedom (no rotation or translation).

Force calculation

The vertical force applied by the rider to the pedal is: $F = m \cdot g = 70 \text{ kg} \times 9.81 \text{ m/s}^2 = 686.7 \text{ N}$.

This force acts downward on the pedal so In the FEA model it is applied as a bearing load distributed over the inner surface of the pedal hole. The opposite side of the pedal is constrained with a fixed support to simulate the pedal axle being clamped to the crank arm.

Moment calculation

The pedal force acting at a distance $d = 0.17$ m from the crank axis causes a bending moment at the crank. The moment is transferred through the crank arm to the bottom bracket. $M = F \times d = 686.7 \text{ N} \times 0.17 \text{ m} = 116.739 \text{ Nm}$.

Application in the FEA model

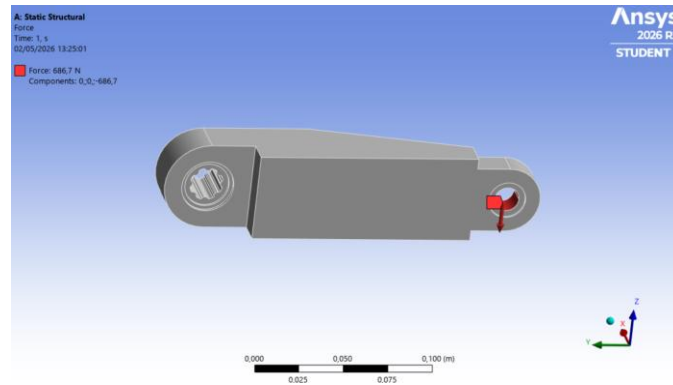


Figure 5 : pedal force (686.7 N) on the inner part of the pedal hole.

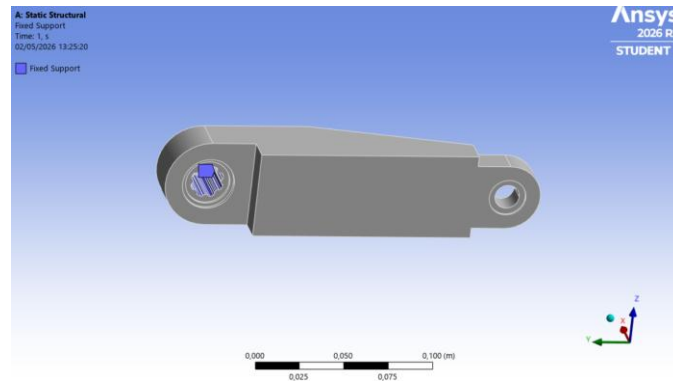


Figure 6 : Fixed support applied to the end of the pedal.

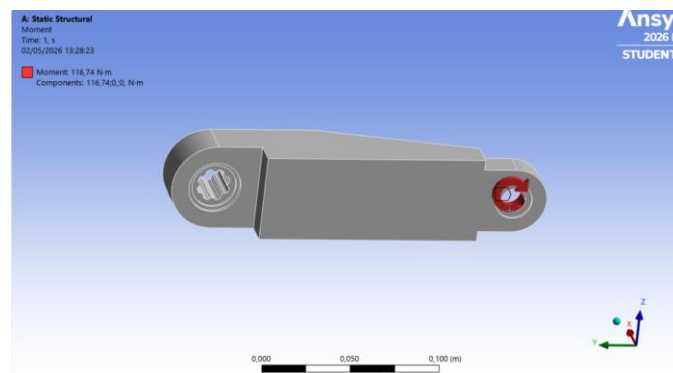


Figure 7 : Bending moment (116.739 Nm).

4. Stage 2

Simulation Setup

The finite element analysis was performed using ANSYS Workbench

Material: Al 7075.

Meshing: A mesh was generated with a refinement for maximum accuracy and lowest computational cost.

Boundary conditions: A bearing load of 686.7 N on the pedal hole a fixed support at the pedal end and a fixed support at the crank shaft end.

Mesh Convergence

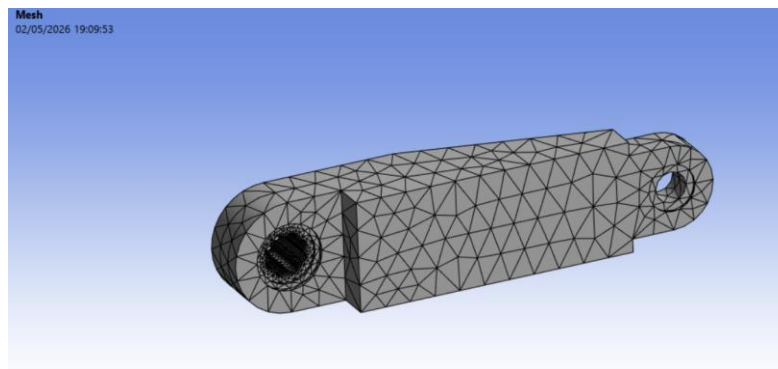


Figure 8 : Baseline design mesh.

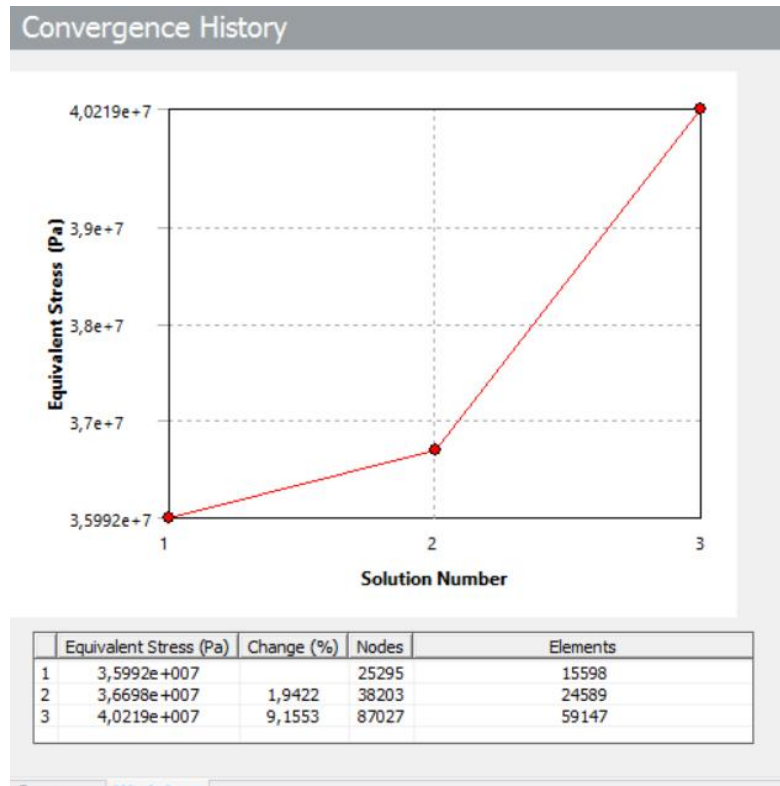


Figure 9 : Mesh convergence study for the baseline design.

Baseline FEA results

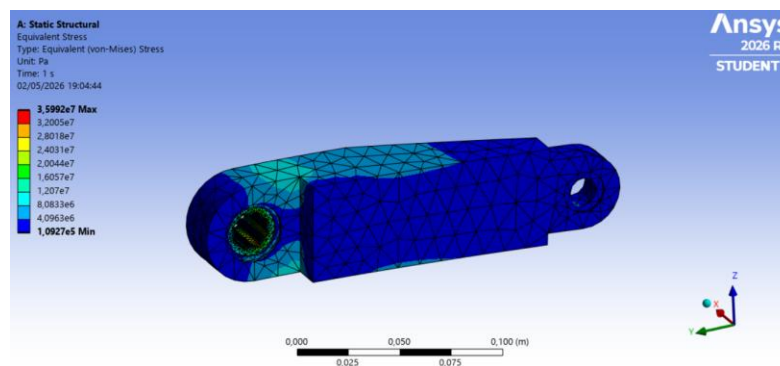


Figure 10 : Equivalent (von Mises) stress (max 35.992 MPa).

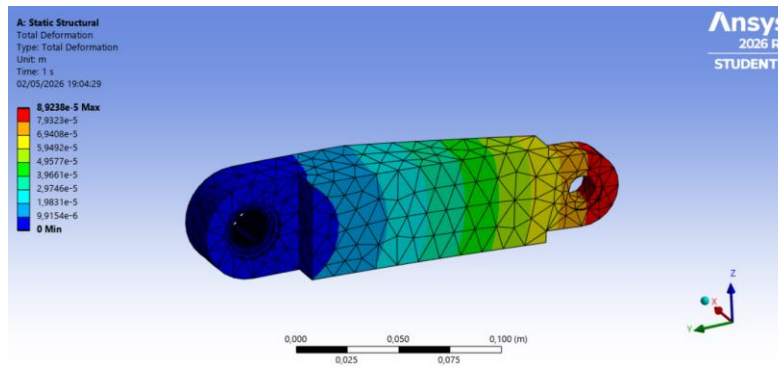


Figure 11 : Total deformation (max 8.9238×10^{-5} m).

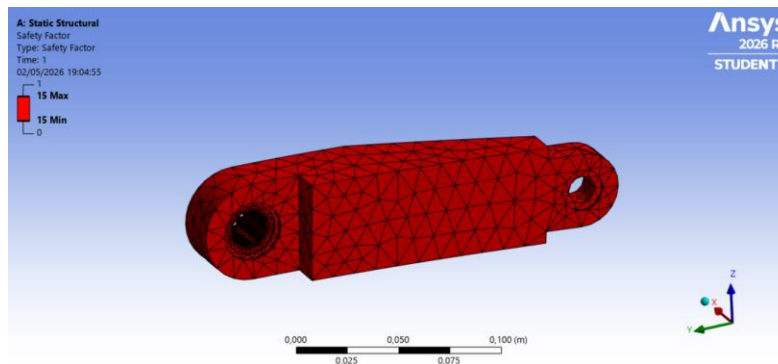


Figure 12 : Safety factor.

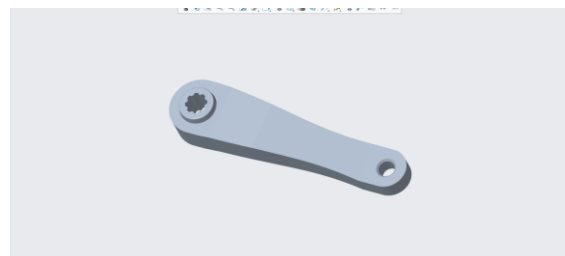
1. Design Optimization

The optimized design has the same fixed interfaces and external envelope but contains material reductions on the back side which follows an I beam crosssection with a thinned web and large central cutouts. The mass of the optimized design is 297 g which is below the 300 g requirement.

The optimized model was meshed and has the same boundary conditions (686.7 N pedal force, pedal fixed, crank shaft fixed).



Front view



Back view (material removed)

Figure 13 : Optimised CAD design: (a) front view, (b) back view showing cut out.

VOLUME = 1.0979897e+05 MM³
 SURFACE AREA = 3.3145610e+04 MM²
 DENSITY = 2.7100000e-06 KILOGRAM / MM³
 MASS = 2.9755521e-01 KILOGRAM

Figure 14 : Optimised design mass from Creo (297 g, meeting the <300 g requirement).

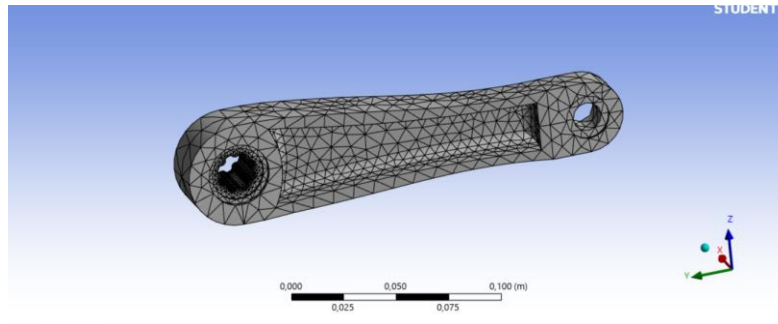


Figure 15 : Optimised design mesh

Mesh Convergence

A convergence check was also performed for the optimised model comparing two mesh refinements. The stress increased by 9%.

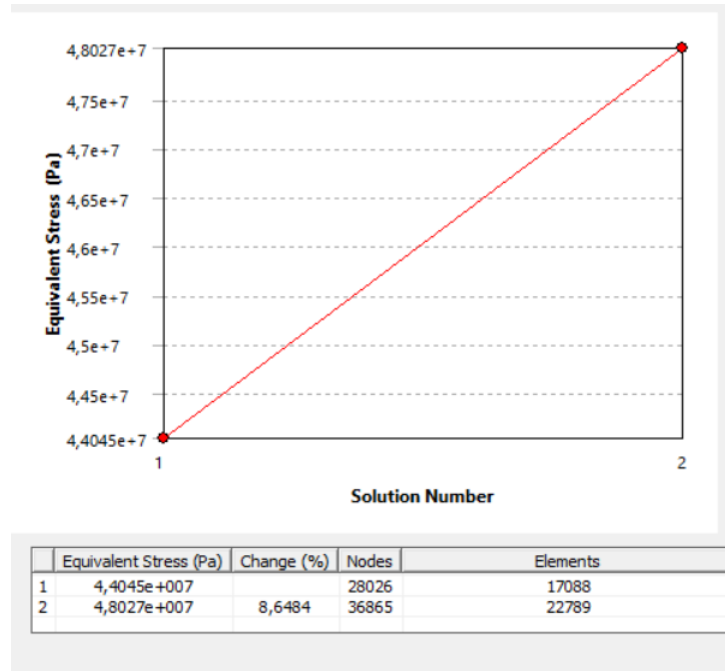


Figure 16 : Mesh convergence for the optimized design.

Optimized FEA results

The maximum stress is 48.027 MPa which is below the yield strength (498 MPa). There is a maximum total deformation of 0.00028001 m (0.28 mm). The safety factor minimum value is 10.37 ($498/48.02 = 10.37$) which is above the required FoS = 2. This means the optimized crank arm is very safe and meets all structural requirements.

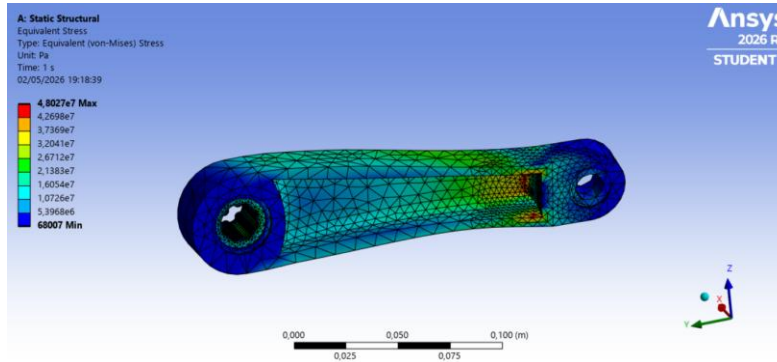


Figure 17 : Optimized design Equivalent stress (max 48.027 MPa).

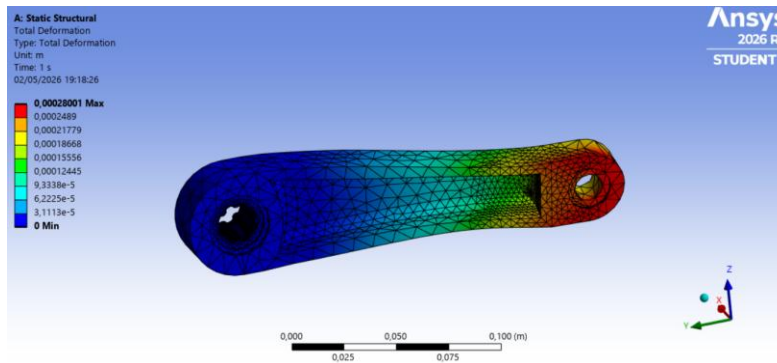


Figure 18 : Optimized design Total deformation (max 0.00028001 m).

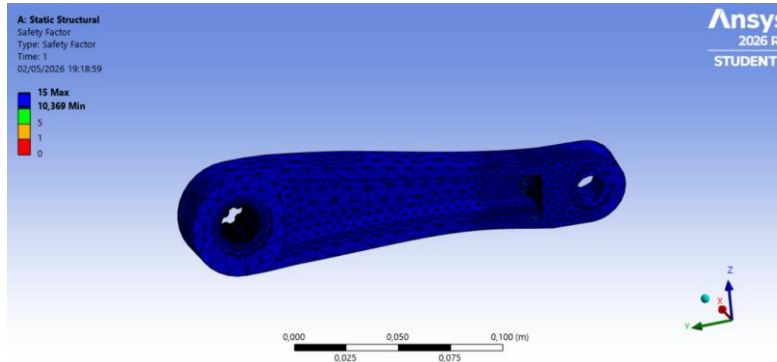


Figure 19 : Optimised design Safety factor.

5. Stage 3 Evaluation

1. Results and Comparison

The baseline design is structurally safe ($FoS = 13.84$) but was too heavy (727 g). The optimized design reduces mass by 59% (727 g to 297 g) while maintaining a safety factor of 10.37. The maximum stress in the optimized design (48.027 MPa) is 9.6% of the yield strength and the deformation is 0.28 mm. The optimization successfully meets the mass target without compromising safety.

2. Specification

Table 3 evaluates the optimized design against each client requirements.

3. Reflection

Mass and strength: By removing material from the back side and creating a thin I beam, mass was reduced to 297 g while achieving a safety factor of 10.37.

Fixed interface regions: The red zones from the model were kept the same so the client would not have any problems applying the crank to the bicycle which would make the model useless.

Manufacturing considerations: The optimized model is suitable for CNC machines. Sharp internal corners were avoided to reduce stress concentrations.

Limitations and remaining risks: The model static load of 686.7 N. But someone riding a BMX would mean dynamic loads (landing from jumps) could be 2x times higher. However since the safety factor is 10.37 it provides a large force allowance.

Table 3 : Validation of optimized design against client specification

Parameter	Requirement	Achieved Yes/No	Evidence
Performance	Client weight 70 kg; Max weight <300 g	Yes (297 g)	Figure 15
Size	Max footprint detailed in Fig. 2	Yes	Engineering drawing
Maintenance	No maintenance during lifespan	Yes	Fatigue strength 147 MPa max stress 48.027 MPa
Finish	Corrosion resistant	Yes	Al 7075 excellent water resistance
Materials	Easily recyclable	Yes	Aluminium is highly recyclable
Transportable	Not easily damaged by impact	Yes	$FoS = 10.37$ provides large impact tolerance

Aesthetics	High-end performance image	Yes	Sculpted I beam with smooth front face
Customer	BMX enthusiasts	Yes	Meets all performance and weight targets
Safety	No sharp protrusions; $FoS \geq 2$	Yes	Minimum $FoS = 10.37$, smooth edges
Environment	$-10\text{ }^{\circ}\text{C}$ to $+40\text{ }^{\circ}\text{C}$, water/dust resistant	Yes	Al 7075 stable in this range
Ergonomics	Compatible with existing mounts/pedals	Yes	Fixed interfaces unchanged

6. Engineering Drawing

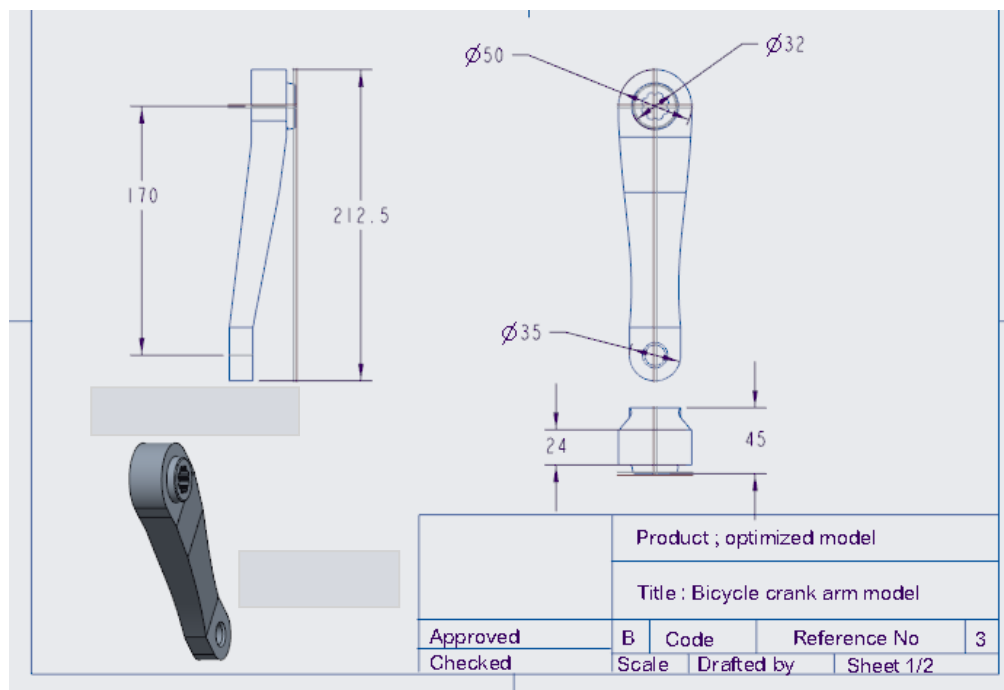


Figure 20 : Final engineering drawing.

7. Conclusion

This report has followed Archer's design process to develop a optimized bicycle crank arm.

By removing material from the back side and creating central cutouts the mass was reduced by 59% to 297 g.

The optimized design was placed under the same loading (686.7 N). The maximum von Mises stress increased slightly to 48.027 MPa which is 9.6% of the yield strength and the safety factor remained at 10.37.

All client specifications have been met (corrosion resistance, recyclability, ergonomics and environmental durability).